

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

CM2 - Economic Modelling (Paper A)

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) A reinsurer covers losses that exceed a pre-defined limit. It is an arrangement, meaning the reinsurer's involvement is triggered only when losses surpass a certain threshold. Which type of reinsurance is it?

1. Quota Share Reinsurance
2. Surplus Reinsurance
3. Facultative Reinsurance
4. Excess of Loss Reinsurance
5. Treaty Reinsurance

[2]

Q.2) What is the correct definition for Gamma in “Greeks”?

1. $= \frac{\partial f}{\partial S}$ = rate of change of the derivative's value with the change in share price
2. $= \frac{\partial^2 f}{\partial S^2}$ = rate of change of Δ with the share price
3. $= \frac{\partial f}{\partial t}$ = rate of change of the derivative's value with time
4. $= \frac{\partial f}{\partial \sigma}$ = rate of change of the derivative's value with volatility
5. None of the above

These assume all other parameters are kept constant.

[2]

Q.3) Each individual is offered the chance to gamble for free. The outcomes of the gamble are distributed as follows: ₹100 gain (i.e. increase in wealth) with probability 20%, no change in wealth with probability 70%, and ₹200 loss with probability 10%. Calculate the expected change in wealth from the gamble.

1. 0
2. 40
3. 20
4. 10
5. -20

[2]

Q.4) Which of the following best defines adverse selection?

1. When insurers charge different premiums for different customers
2. When healthy individuals opt out of insurance, leaving a pool of high-risk individuals
3. When all customers are charged the same premium
4. When insurers deny claims due to non-disclosure
5. When fraud is committed by policyholders

[2]

Q.5) Which of the following is not a key assumption of the Capital Asset Pricing Model (CAPM)?

1. All investors are rational and risk-averse
2. There is a risk-free rate at which investors can borrow or lend money
3. Investors hold diversified portfolios to eliminate unsystematic risk
4. Investors can only invest in individual assets, not in portfolios
5. All of the above

[2]

Q.6) The primary benefit of diversification in a portfolio is to:

1. Increase the overall return of the portfolio
2. Eliminate the risk of individual assets
3. Reduce the portfolio's overall risk without sacrificing expected return
4. Focus investments on high-volatility assets for higher returns
5. Investing exclusively in high-growth stocks for greater returns

[2]

Q.7) In the context of utility maximization, individuals are assumed to:

1. Always aim to maximize their wealth, regardless of satisfaction.
2. Allocate their income in such a way that the marginal utility per dollar spent is equal across all goods.
3. Maximize utility by focusing on a single good.
4. Spend all their income on the good that provides the highest total utility.
5. Spend all income on a single good that provides maximum satisfaction.

[2]

Q.8) In CAPM, if the market portfolio has a beta of 1 and a risk-free security has a beta of 0, how do their expected returns compare?

1. The market portfolio has a higher expected return than the risk-free security.
2. The expected returns are the same.
3. The risk-free security has a higher expected return than the market portfolio.
4. The market portfolio's expected return is negative.
5. Highly correlated with the market's movements.

[2]

Q.9) Which of the following factors does not affect the price of a call option?

1. The current price of the underlying asset.
2. The time to expiration of the option.
3. The strike price of the option.
4. The inflation rate.
5. The premium paid for the option.

[2]

Q.10) For a European call option, the value at expiration when the underlying asset price is above the strike price is:

1. Zero.
2. The strike price.
3. The difference between the underlying asset price and the strike price.
4. The total premium paid for the option.
5. Null Value

[2]

Q.11) In the Cameron-Martin-Girsanov Theorem, the original process, say X_t , transforms under a new measure into a process Y_t such that:

1. The process remains the same but its variance increases.
2. The drift of the process changes but the diffusion coefficient remains the same.
3. The process changes only in its volatility.
4. Both the drift and the volatility change in the transformation.

5. Changing the drift term in a stochastic differential equation

[2]

Q.12) In the classical Lundberg risk model, which of the following statements is true?

1. Claims arrive according to a deterministic process and are of constant size.
2. The premium income increases linearly over time, and claim sizes are normally distributed.
3. Claims arrive according to a Poisson process, and claim sizes are i.i.d. random variables.
4. The surplus process decreases linearly over time in the absence of claims.
5. The model assumes there is no initial surplus in the insurer's reserve.

[2]

Q.13) Which of the following would NOT be an example of diminishing marginal utility?

1. The satisfaction from drinking water decreases as more glasses are consumed on a hot day.
2. A shopper becomes less excited about purchasing clothes after buying several items.
3. A person enjoys the first few bites of chocolate but finds the last few bites less enjoyable.
4. A person becomes more satisfied with their income as they earn higher wages.
5. A person feels increasingly satisfied with each new car purchase.

[2]

Q.14) A portfolio consists of two types of policies. For type 1, the number of claims in a year has a Poisson distribution with mean 1.5 and the claim sizes are exponentially distributed with mean 5. For type 2, the number of claims in a year has a Poisson distribution with mean 2 and the claim sizes are exponentially distributed with mean 4. Let S be the total amount claimed on the whole portfolio in one year. All policies are assumed to be independent.

Determine the mean and variance of S .

1. $E(S) = 20$; $\text{Var}(S) = 254$
2. $E(S) = 15.5$; $\text{Var}(S) = 176$
3. $E(S) = 15$; $\text{Var}(S) = 345$
4. $E(S) = 14.8$; $\text{Var}(S) = 154$
5. $E(S) = 15.5$; $\text{Var}(S) = 139$

[2]

Q.15) Which of the following is not the characteristics of a short-term insurance contract?

- A. Policy lasts for a fixed term
- B. Policy lasts for a relatively short period of time
- C. Insurer pays claims that arise during the policy term
- D. Option (but no obligation) to renew policy
- E. Claim bring policy to an end

1. E
2. D
3. A
4. B
5. C

[2]

Q.16) Claims under a particular class of insurance follow a lognormal distribution with mean 9.070 and standard deviation of 10.132 (figures in ₹000s). In any one year 20% of policies are expected to give rise to a claim. An insurance company has 200 policies on its books and wishes to take out individual excess of loss reinsurance to cover all the policies in the portfolio.

The reinsurer has quoted premiums for two levels of reinsurance as follows (figures in ₹000s)

Retention Limit	Premium
25	48.5
30	38.2

Calculate the probability, under each reinsurance arrangement, that a claim arising will involve the reinsurer.

1. Retention Limit: 25, Probability: 0.0756; Retention Limit: 30, Probability: 0.0483
2. Retention Limit: 25, Probability: 0.0849; Retention Limit: 30, Probability: 0.0872
3. Retention Limit: 25, Probability: 0.0574; Retention Limit: 30, Probability: 0.0376
4. Retention Limit: 25, Probability: 0.0745; Retention Limit: 30, Probability: 0.0278
5. Retention Limit: 25, Probability: 0.0228; Retention Limit: 30, Probability: 0.0983

[3]

Q.17) An investment analyst assumes the return on a fund follows a discrete distribution as set out in the table below:

Return (% p.a.)	Probability (%)
0	10
2	20
4	40
6	25
10	5

Calculate the expected return and the variance of the return.

1. $E(X) = 10\%$; $\text{Var}(X) = 0.055\%$
2. $E(X) = 4\%$; $\text{Var}(X) = 0.052\%$
3. $E(X) = 7\%$; $\text{Var}(X) = 0.076\%$
4. $E(X) = 44\%$; $\text{Var}(X) = 0.098\%$
5. $E(X) = 54\%$; $\text{Var}(X) = 0.043\%$

[3]

Q.18) Using data in Q17 above, Calculate the following risk measures:

- (a) The semi-variance of return
- (b) The shortfall probability relative to a benchmark return of 5%
- (c) The expected shortfall relative to a benchmark return of 5%.

1. (a) 0.056%; (b) 50%; (c) 1%
2. (a) 0.024%; (b) 70%; (c) 1.5%
3. (a) 0.078%; (b) 78%; (c) 8%
4. (a) 0.054%; (b) 30%; (c) 7.5%
5. (a) 0.034%; (b) 20%; (c) 6.2%

[3]

Q.19) Consider two individuals with the following utility functions:

- Individual A: $U_A(w) = w + 0.01w^2$, where $w > 0$
- Individual B: $U_B(w) = \ln(w)$, where $w > 0$

Both individuals have a current wealth of ₹300.

A gamble is offered with a 50% chance of increasing their wealth to ₹600 and a 50% chance of reducing it to ₹150.

Which individual is more likely to accept the gamble based on expected utility theory?

1. Individual A is more likely to accept the gamble.
2. Individual B is more likely to accept the gamble.
3. Both individuals will accept the gamble.
4. Both individuals will reject the gamble.
5. Not enough information to determine.

[3]

Q.20) The loss severity distribution for a portfolio of household insurance policies is assumed to be Pareto with parameters $\alpha = 3.5$, $\theta = 1,000$. Next year, losses are expected to increase by 5%, and the insurer has decided to introduce a policyholder excess of ₹100.

Calculate the probability that a loss next year is borne entirely by the policyholder.

1. 0.5786
2. 0.2727
3. 0.3563
4. 0.7543
5. 0.6588

[3]

Q.21) A firm owns a warehouse worth ₹1,000,000. It can either maintain it properly or cut corners:

- Proper maintenance reduces fire probability to 1%
- Cutting corners increases fire probability to 5%
- Maintenance costs ₹10,000/year

The firm has full fire insurance with a premium P . If moral hazard is a concern, what is the maximum premium P that still incentivizes the firm to maintain the building?

1. ₹40,000
2. ₹45,000
3. ₹50,000
4. ₹55,000
5. ₹60,000

[3]

Q.22) An investor buys, for a premium of 187.06, a call option on a non-dividend paying stock whose current price is 5,000. The strike price of the call is 5,250 and the time to expiry is 6 months. The risk-free rate of return over this period is 5% p.a. The Black-Scholes price for a call option on a non-dividend paying share is:

$$c = S_t \Phi(d_1) - Ke^{-r(T-t)} \Phi(d_2)$$

$$\text{where } d_1 = \frac{\log S_t / K + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}$$

$$d_2 = d_1 - \sigma\sqrt{T-t}$$

and S_t = current price σ = volatility $T-t$ = time to expiry
 K = strike price r = risk free rate

Calculate the price of a put option with the same time to maturity and strike price as the call.

1. 435.76
2. 307.44
3. 560.43
4. 376.98
5. 437.23

[3]

Q.23) An individual faces a 10% chance of incurring a ₹10,000 loss due to an accident. If uninsured, they take precaution costing ₹200, which reduces the chance of loss from 10% to 5%. If fully insured (insurance pays the entire ₹10,000 if a loss occurs), the individual stops taking the precaution.

What is the increase in the insurer's expected payout due to moral hazard?

1. ₹200
2. ₹300
3. ₹400
4. ₹500
5. ₹600

[3]

Q.24) You hold a portfolio with:

- Stock A: 40% weight, $\beta = 1.5$
- Stock B: 60% weight, $\beta = 0.75$

Market risk premium = 6%, risk-free rate = 2%.

What is the expected return of the portfolio using CAPM?

1. 6.5%
2. 7.0%
3. 7.5%
4. 8.3%
5. 9.0%

[3]

Q.25) The loss amount, X , from a portfolio of non-life insurance policies is assumed to be independently distributed with mean ₹800 and standard deviation ₹1,200.

Calculate the values of the parameters of a lognormal distribution with this mean and standard deviation.

1. $\text{Sigma}^2 = 1.178655$; $\mu = 5.674533$
2. $\text{Sigma}^2 = 1.876590$; $\mu = 6.095284$
3. $\text{Sigma}^2 = 1.994583$; $\mu = 5.674533$
4. $\text{Sigma}^2 = 1.178655$; $\mu = 6.095284$
5. $\text{Sigma}^2 = 1.342579$; $\mu = 5.674533$

[3]

Q.26) The tables below show the cumulative incurred claims data and number of reported claims, by accident year and development year (in ₹'000s).

Cumulative incurred claims

Accident Year	Development Year				
	0	1	2	3	Ult
2000	3,417	4,291	4,581	4,714	4,900
2001	4,814	6,888	7,007		
2002	5,844	8,000			
2003	6,654				

Cumulative number of reported claims

Accident Year	Development Year				
	0	1	2	3	Ult
2000	414	460	482	488	500
2001	453	506	526		
2002	496	558			
2003	540				

Estimate the ultimate number of claims, for each accident year, using the chain-ladder technique. (in ₹000s)

1. 500, 657.4, 676.8, 654.8
2. 500, 565.5, 603.7, 655.9
3. 500, 545.6, 604.0, 653.6
4. 500, 554.7, 675.3, 665.7
5. 500, 598.6, 675.7, 699.3

[4]

Q.27) Basis information given in Q.26 above, Estimate the ultimate average incurred cost per claim, for each accident year, using the grossing-up method.

1. 9.800, 13.736, 14.758, 15.312
2. 9.540, 14.636, 15.143, 14.876
3. 9.640, 13.763, 14.758, 15.009
4. 9.450, 14.234, 16.657, 14.980
5. 9.890, 13.777, 15.654, 15.867

[4]

Q.28) Basis information given in Q.26 above and using the results from Q.26 and Q.27 above, calculate the total reserve required, assuming that claims paid to date are ₹19,212,000.

1. ₹12,104,000
2. ₹12,104,500
3. ₹14,760,000
4. ₹14,756,900
5. ₹13,870,000

[4]

Q.29) An investor is trying to choose between the investments whose distributions of returns are described below:

Investment A: 0.4 probability that it will return 10%
0.2 probability that it will return 15%

0.4 probability that it will return 20%

Investment B: 0.25 probability that it will return 10%
0.70 probability that it will return 15%
0.05 probability that it will return 40%

Investment C: A uniform distribution on the range 10% to 20%

Calculate the following for each investment:

- (i) expected return
- (ii) variance of return

1. A:10%,B:8%,C:11%; A:4.33%,B:6%,C:4.8%
2. A:15%,B:8%,C:11%; A:.02%,B:6%,C:4.8%
3. A:10%,B:15%,C:11%; A:4.33%,B:.375%,C:4.8%
4. A:10%,B:8%,C:15%; A:4.33%,B:6%,C:.0833%
5. A:15%,B:15%,C:15%; A:.02%,B:.375%,C:.0833%

[4]

Q.30) Using information given in Q.29, Calculate the following for each investment:

- (i) semi-variance
- (ii) shortfall probability below 15%

1. A:.10%,B:.0625%,C:.0417%; A:.0.4,B:0.25,C:0.5
2. A:.50%,B:.0625%,C:.0417%; A:0.2,B:0.5,C:0.2
3. A:.10%,B:.0625%,C:.0298%; A:0.8,B:0.5,C:0.6
4. A:.10%,B:.0543%,C:.0417%; A:0.8,B:0.5,C:0.7
5. A:.80%,B:.0625%,C:.0417%; A:0.9,B:0.5,C:0.2

[4]

Q.31) Loss amounts under a class of insurance policies follow an exponential distribution with mean 100. The insurance company wishes to enter into an individual excess of loss reinsurance arrangement with retention level M set such that 8 out of 10 claims will not involve the reinsurer.

Find the retention M.

1. 160.94
2. 123.87
3. 198.87
4. 224.65
5. 176.45

[4]

Q.32) Assets A and B have the following distributions of returns in various states:

State	Asset A	Asset B	Probability
1	10%	-12%	0.1
2	8%	0%	0.2
3	6%	3%	0.3
4	4%	16%	0.4

Calculate the correlation coefficient between the returns on asset A and asset B.

1. -0.970
2. +0.970
3. -0.874
4. +0.874
5. None of the above

[4]

Q.33) A portfolio of insurance policies has two types of claims:

- Loss amounts for Type I claims are exponentially distributed with mean 120.
- Loss amounts for Type II claims are exponentially distributed with mean 110.

25% of claims are Type I, and 75% are Type II.

Calculate the mean and variance of the loss amount for a randomly chosen claim.

1. Mean = 97.4; Variance = 12,694
2. Mean = 165.4; Variance = 13,492
3. Mean = 112.5; Variance = 13,492
4. Mean = 112.5; Variance = 12,694
5. Mean = 97.4; Variance = 13,492

[4]

Q.34) An insurer believes that claims from a particular type of policy follow a Pareto distribution with parameters = 2.5 and = 300. The insurer wishes to introduce a deductible such that 25% of losses result in no claim on the insurer.

Calculate the size of the deductible.

1. 55.45
2. 98.62
3. 36.59
4. 66.87
5. 78.53

[4]

Q.35) A stock is currently trading at ₹100. A European call option on the stock has a strike price of ₹105 and expires in 6 months. The risk-free interest rate is 5% per annum (continuously compounded), and the volatility (σ) of the stock is 20% per annum.

Using the Black-Scholes formula, what is the price of the call option?
(Use standard normal values: $N(d_1) = 0.5379$, $N(d_2) = 0.4990$)

1. ₹4.12
2. ₹5.03
3. ₹2.76
4. ₹6.11
5. ₹8.43

[4]
